

5 (a) (i) X-ray photons produced when electrons decelerate [1]

Photon energy (and hence wavelength) varies continuously since electrons decelerate across a continuous range / lose a continuous range of KE [1]

(ii) Electrons in the inner shell / lower energy level are knocked out and outer shell electrons de-excite causing an emission of a photon [1]
Energy levels are discrete and therefore, energy, and hence wavelength, of photon emitted are discrete [1]

(iii) $eV = hc / \lambda$ [1]
 $(1.6 \times 10^{-19})(15000) = (6.63 \times 10^{-34})(3.0 \times 10^8) / \lambda$ [1]
 $\lambda = 8.3 \times 10^{-11} \text{ m}$ [1]

(iv) lower $\lambda_{\min}$ [1]
 Higher intensity everywhere else and same characteristic λ [1]

(b) (i) $\Delta x \Delta p \geq h$ [1]
 $(1.00 \times 10^{-10})(\Delta p) \geq h$ [1]
 $\Delta p = 6.63 \times 10^{-24} \text{ kg m s}^{-1}$ [1]

(ii) Electrons are likely to be scattered through a significant angle **OR**
 Emerging electrons will be travelling in a range of directions [1]

(c) (i) **classical explanation:** intensity proportional to wave amplitude-squared **or**
 intensity is energy delivered per second per unit area of wave front [1]

(accept depends on / determined by amplitude, but not proportional)

quantum explanation: intensity proportional to the rate of arrival of photons or photons per second [1]

(ii) **classical explanation:** continuous absorption of energy from wave [1]
quantum explanation: absorption in quanta or photons [1]

(d) (i) x-intercept = 1.93×10^6 [1]

$f_0 = c / \lambda_0 = (c)(1 / \lambda_0) = (3.0 \times 10^8)(1.93 \times 10^6) = 5.79 \times 10^{14}$ [1]

(ii) correct line [1]



